

Customer Churn Analytics Project Report

Project Overview

This project analyzes customer purchasing behavior, revenue performance, and churn using SQL and Power BI. The goal is to identify churn drivers, monitor KPIs, and recommend strategies to improve customer retention.

Business Problem

Customer churn reduces long-term revenue and customer lifetime value. The dashboard helps identify high-risk customer segments and revenue at risk.

Objectives

- Analyze customer behavior.
- Measure churn and retention.
- Build interactive Power BI dashboards.
- Provide actionable business recommendations.

Dataset

The project uses Customers, Orders, and Products tables containing customer demographics, transactions, product information, membership tiers, acquisition channels, and churn indicators.

Tools Used

MySQL, Power BI, Microsoft Excel, GitHub.

Database Model

Relational model with Customers → Orders ← Products using primary and foreign keys.

Data Cleaning

Trimmed text, standardized values, validated relationships, handled duplicates and missing values, and verified data quality.

KPIs

Total Customers, Total Orders, Total Revenue, Churn Rate, Retention Rate, Average Order Value, Revenue at Risk, Return Rate, Customer Lifetime Value.

Key Business Insights

Electronics generated the highest revenue. Organic Search contributed the highest acquisition revenue. Gold members showed relatively high churn despite strong revenue contribution. Customers inactive for more than 90 days had the highest churn risk.

Conclusion

The project demonstrates end-to-end data analytics skills including SQL querying, data cleaning, data modeling, KPI development, dashboard design, and business storytelling.